

ADJUSTABLE LED LIGHT INTENSITY CONTROL

1. Aim

To construct a simple electronic setup in which the light intensity of an LED can be varied manually using a potentiometer and to study the effect of changing electrical resistance on LED illumination.

2. Problem Statement

LEDs are commonly used as indicators and lighting elements, but in many situations the required light intensity may not remain the same.

A simple method is therefore needed to increase or decrease the amount of light produced by an LED according to the requirement.

This project demonstrates an easy manual method of obtaining different levels of LED illumination.

3. Proposed Solution

An LED circuit is assembled on a breadboard along with a potentiometer and a suitable resistor.

The potentiometer is adjusted manually to obtain different LED light levels. The change in the circuit resistance affects the electrical current available to the LED, producing a visible change in its intensity.

This provides a simple demonstration of electrical control through a variable component.

4. Components Required

LED

Potentiometer

Current limiting resistor

Breadboard

9V battery

Connecting wires

5. Principle of Operation

The project works based on the relationship between resistance and current in an electrical circuit.

The potentiometer provides an adjustable resistance. By changing its position, the electrical conditions of the LED circuit are changed.

As the potentiometer is adjusted, the LED changes from a brighter state to a dimmer state or vice versa.

The resistor provides additional protection by restricting excessive current through the LED.

6. Construction Procedure

1. Take a breadboard and place the LED on it.
2. Connect the LED terminals to the appropriate breadboard rows.
3. Insert the resistor in the LED circuit.
4. Connect the potentiometer to the circuit.

5. Connect the battery using jumper wires.
6. Make sure the positive and negative terminals are connected correctly.
7. Inspect the circuit for loose or incorrect connections.
8. Switch on the battery supply.
9. Slowly turn the potentiometer.
10. Note the different illumination levels obtained from the LED.

7. Applications

Display Indicators – Controlling the intensity of indicator lights.

Miniature Lighting Systems – Adjusting light output in small circuits.

Electronic Learning Kits – Demonstrating practical circuit concepts.

Decorative Electronics – Producing different lighting effects.

Control Panels – Useful as a basic example of manual intensity control.

Prototype Development – Helps test LED behaviour under different circuit conditions.

8. Advantages

Requires only a few easily available components.

Simple breadboard construction.

Does not require programming.

Provides immediate visual output.

Easy to operate and demonstrate.

Helps understand practical electrical concepts.

Can be modified for different LED applications.

Suitable for basic electronics experiments.

9. Result

The LED control system was assembled successfully on a breadboard.

Different levels of LED illumination were obtained by adjusting the potentiometer. The experiment successfully demonstrated that a variable electrical input can be used to control the visible output of an LED.

10. Conclusion

The project successfully demonstrates a basic manual LED intensity control system.

By combining a potentiometer with an LED circuit, different levels of illumination can be obtained without using a microcontroller or programming.

The experiment provides practical exposure to electrical resistance, current control, circuit connections and LED characteristics.

11. Project Demonstration

Project Video / Demo:
[Add Video Link Here]

12. Circuit Diagram

[Add Circuit Diagram Image Here]

Circuit Flow:

9V Battery → Potentiometer → Resistor → LED → Battery

The potentiometer is adjusted to obtain the required LED illumination.

13. Project Images

A. Breadboard Setup

B. LED in Working Condition

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B. LED in Working Condition

[Insert your LED glowing photo here]

C. Potentiometer Adjustment

[Insert photo showing potentiometer being adjusted here]

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